Big Data: The Ethics of the Information Economy

This project will take a dive into the ethics of how large corporations and businesses use consumer data (information about you and I) to make informed business decisions. Although big data drives innovation, economic growth, and competitive advantages in the information economy, it is increasingly being used in ways that violate privacy, exploit individuals, reinforce inequality, and strip ordinary people of control over their own digital identities.

Project Significance and Background

The importance of this research project lies in its examination of big data, one of the most powerful drivers of today's information economy. It has increasingly been used in ways that harm, exploit, or disadvantage regular people, and while it is often praised for increasing innovation, informing decision making, and improving organization. The benefits to this data mining tend to accumulate to institutions, corporations, and governments rather than the people who provide this data, and I'd argue against the ethics of this. Ordinary people frequently do not understand how their data is collected, who profits from it, or how algorithmic systems use it to make decisions that directly affect their lives. As a result, big data contributes to a widening imbalance of power, where individual's information is stored in databases somewhere, and the organizations that use this data are ever unclear about how they are using it. Investigating these dynamics is crucial because data-driven decision making now shapes access to jobs, healthcare, loans, education, housing, and even political information. Understanding the ethical risks is essential to protecting the agency of individuals and ensuring accountability for these firms in an increasingly data dependent society. This project specifically will be guided by two main theoretical frameworks. First, data ethics will serve as a foundation for analyzing how personal data should be collected, used, shared, interpreted, and how the user should be informed. It also will examine issues including privacy, consent, fairness, and the moral responsibilities of institutions that develop or deploy data-driven technologies. This framework helps clarify why certain uses of big data are not merely problematic, but fundamentally unethical because they violate individuals' dignity and right to self-determination.

Second, the project will critically assess the politics of the situation to examine how big data functions within larger systems of economic incentives and power. This perspective is crucial to understand how the information economy treats personal data as a commodity and encourages companies to collect as much as possible (even without *meaningful* consent) because of its value for prediction, targeting, and profit. By understanding big data within how it functions in our free market, the project shows how unethical practices are not accidental but are incentivized by that market, and how the power creep of data impacts these companies and their business practices.

Together, these frameworks reveal not only how big data harms regular people, but also why these harms persist and what ethical questions these companies need to ask themselves in order to address them.

Research Methods

To complete this project on the unethical uses of big data toward regular people, I will follow a structured research process that combines scholarly research, critical analysis, and synthesis of real-world examples. Each step is necessary to ensure the project is grounded in credible evidence, guided by strong theory, and supported by concrete cases of how big data affects individuals in everyday life.

Step 1: Conduct a comprehensive literature review (Nov–Dec).
I will begin by reviewing peer-reviewed academic articles, books, and policy reports on data ethics, surveillance, algorithmic bias, digital privacy, and the political economy of big data. This step is essential because it establishes context for the issue, identifies existing discussions on the topic, and helps me define what I'm focusing on for research. Reviewing perspectives that support and oppose the current use of big data will allow me to frame the problem as a tension between economic/business benefit and ethical harm.

Step 2: Analyze critical texts, case studies, and real-world incidents (Jan).
Next, I will identify and study specific examples where big data systems have harmed regular individuals—such as algorithmic discrimination in hiring, biased predictive policing, targeted political manipulation, discriminatory insurance pricing, or data breaches affecting millions. Analyzing these cases is necessary to move beyond theory and show how unethical data practices materially impact people's lives. These examples will also help illustrate broader issues that are connected to data ethics, politics, and economics.

Step 3: Apply theory to the findings (late Jan-Feb).
Using data ethics and critical political economy as guiding frameworks, I will interpret the patterns uncovered during the reviews and case analyses. This step is crucial because it allows me to explain not only what's wrong, but also why it's occurring and how they connect to larger systems of power, profit, and digital surveillance. The frameworks will help organize the project's argument and ensure it remains grounded in theory and research.

Step 4: Draft, revise, and finalize the research project (Feb-March).
In the final stage, I will synthesize my findings into a well-structured written paper. Revision is necessary to ensure clarity, coherence, and strong argumentation, and the project will be completed by late March or early April, with final edits made before the end of the Spring semester.

Expected Outcome/Deliverables

Upon completing this research, the primary deliverable will be a comprehensive academic paper detailing the findings from the literature review and case study analyses. The paper will be structured and suitable for submission to a scholarly journal that specializes in data ethics, critical media studies, or the sociology of technology.

The products of this research will be shared through two distinct channels:
Firstly, I will seek to present the findings at an academic conference, such as the UCF Student Scholar Symposium or a relevant research conference for undergraduates. This will allow the project's insights to be shared, debated, or even expanded upon within an academic community. Second, to ensure the findings reach the average person (who is most affected by these issues), I will create a concise, accessible pamphlet or article for a magazine. This summary will translate the academic frameworks into more practical, actionable, and digestible information for a general audience. I'd also share it with student organization members and officers to promote the ideas of my project, and attempt to make real change within the UCF community.

The new findings of this project will be useful for, again, two reasons:
Firstly, in the field of data ethics, this research will provide a critical analysis of data ethics and political economy frameworks. But, instead of treating these ethics problems or policy failures as if they're in a vacuum, this project will offer a more integrated model. It will demonstrate how the profit-driven incentives of the information economy directly produce ethics problems, and how they do not just come about as a lesser side effect. The new result will be clearer findings of how privacy violations and profit bias function as necessary, profitable features of the current market, rather than as accidental bugs.

For the UCF community, this project will stress the criticalness of data literacy for all. Students, faculty, and staff are constantly navigating data-driven systems, from educational software and social media to job and loan applications. This research will offer a spectacle for understanding how their personal data is being used and monetized, and allow them to come to conclusions about how they want to sell information about themselves, and to whom. The new understanding gained by the community will be a heightened awareness of how abstract concepts of "big data" translate into tangible, real-world impacts on all parts of their lives, empowering them to become more informed and critical digital citizens.

Citations

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Preliminary Work and Experience

My capacity to complete this project is supported by academic, professional,and personal experience. As a student, I'm interacting with concepts and problems from a critical perspective on a daily basis, and this general training is perfectly fitting for a project like this, separate from the courses I've taken on critical thinking, writing, and presentation. As a System Administration Intern at a managed service provider, B2 solutions, I have gained a practical, ground-level perspective on how modern corporations handle data, which directly informs my analysis of business incentives, and how this data helps me as a professional, achieve my goals directly. Furthermore, as a dedicated technology enthusiast, I read the news on current and real-world incidents that assist in my understanding of this topic and others. This combination of technical experience and an interest within the subject provides a solid foundation for writing a paper on the complex intersection of technology, ethics, and political economy that this project demands.

IRB/IACUC statement

This project does not require IRB or IACUC approval, as it involves analysis of published literature, case studies, and publicly available data only, with no direct interaction with human or animal subjects.

Budget

The cost of this research project would be very low. Because the project only uses books, articles, and case studies, there’s no need for lab equipment, surveys, or travel. The only possible costs might be for printing, photocopying, or buying a few books or journal articles, which would be within the $200 range at maximum. Everything else can be done for free using the library, online resources, and a computer.